

Vegetation and the Conditional Distribution of Urban Temperature Anomalies: A Semiparametric Analysis

Lizan Meryl Pereira, Soudeep Deb

Decision Sciences, Indian Institute of Management Bangalore, Bannerghatta Road, Bengaluru, KA 560076, India.

Abstract

Urban vegetation is widely recognized as an important mechanism for moderating land surface temperature, but its association with temperature may vary across vegetation levels and across different parts of the temperature distribution. This study examines the nonlinear and distributional relationship between normalized difference vegetation index (NDVI) and land surface temperature (LST) anomalies in Bengaluru, India, using satellite-derived observations within the BBMP boundary from 2000 to 2025. A semiparametric quantile regression framework with natural cubic splines is used to model the flexible NDVI-LST relationship across the conditional quantiles. The analysis controls for urban category and nighttime lights, and extended specifications incorporate lagged LST and neighboring NDVI to examine temporal persistence and surrounding vegetation associations. The results show that the NDVI-LST relationship is nonlinear and differs substantially across the conditional distribution of LST anomalies. Vegetation loss is more strongly associated with increases in the upper quantiles than in the lower quantiles, indicating that greenness is particularly relevant under hotter surface-temperature conditions. The fitted quantile curves further suggest that higher NDVI is associated not only with lower median LST anomalies, but also with a narrower temperature distribution, mainly through reductions in the upper tail. Neighboring vegetation is also negatively associated with local LST anomalies, with stronger associations at higher quantiles and under more local spatial weighting. These findings highlight the importance of modeling both nonlinear vegetation effects and distributional temperature responses. The results suggest that vegetation-based interventions may be especially useful for reducing high-temperature anomalies and improving thermal stability in rapidly urbanizing cities.

Keywords: Quantile regression; Semiparametric modeling; Natural cubic splines; Remote sensing data; Urban heat island

1. Introduction

Rapid urbanization changes land use and land cover, often replacing vegetation with built-up surfaces that absorb and retain more heat. These changes contribute to the urban heat island (UHI), where urban areas are warmer than their surrounding rural areas. Increased urban heat is a major concern because of its consequences for ecosystems, water availability, and human health, including heat-related and respiratory illnesses (Sussman et al., 2021). Urban green spaces are therefore important for both temperature moderation and public well-being (Kondo et al., 2018). The intensity of urban heat depends on several factors, including vegetation, seasonal conditions, geography, and anthropogenic activity (Kim and Baik, 2005).

Our empirical analysis focuses on Bengaluru, the capital city of Karnataka, India, where urban expansion and the loss of vegetation have raised concerns about increasing surface temperatures. Although the inverse relationship between vegetation and land surface temperature (LST) has been documented in several settings (Sharma et al., 2022; Kliengchuay et al., 2025), understanding how this association varies with vegetation levels is important for assessing the role of the remaining green

cover. Changes in land cover can influence both surface temperatures and the spatial extent of urban heat (Mohammad et al., 2019), making Bengaluru a relevant setting for examining these relationships over time.

Several studies have examined vegetation and temperature using regression and spatial modeling approaches. For example, Tuoku et al. (2024) studied links between the normalized difference vegetation index (NDVI) and climatic factors using multiple regression, while Eshetie (2024) used ordinary least squares (OLS) and geographically weighted regression to examine spatial variation in LST and its associated factors. Temporal relationships have also received attention: Kumar et al. (2023) examined lagged and accumulated vegetation responses to climatic factors, including LST. Other studies have considered distributional and nonlinear relationships. Gu and You (2022) used spatial quantile regression to examine LST and its associated factors across quantiles, and Alavipanah et al. (2015) used kernel quantile regression to study LST and land cover in Munich. In Bengaluru, Shah et al. (2021) examined how the cooling association of urban vegetation varies with distance from green spaces.

These studies provide important evidence, but there remains scope to examine nonlinear, distributional, and neighborhood associations together using a long-term urban dataset. Vegetation is measured here by NDVI, which ranges from -1 to 1 , with higher positive values generally indicating greater greenness. A linear mean regression assigns a constant slope to NDVI and describes its association with average LST. It cannot establish whether the association changes across vegetation levels or differs between the lower and upper parts of the conditional temperature distribution. These distinctions motivate two related questions:

Does the NDVI–LST association vary nonlinearly across the NDVI range, and does its magnitude differ across conditional LST quantiles?

Figure 1 provides an initial illustration of these questions. Panel (a) compares the estimated associations for a 0.1-unit decrease in NDVI from OLS and quantile regression. The quantile estimates increase from approximately 0.17°C at $\tau = 0.10$ to 0.56°C at $\tau = 0.90$, whereas OLS provides a single estimate of approximately 0.36°C . Panel (b) presents the mean and selected quantiles of deseasonalized LST within eight NDVI bins, plotted against the mean NDVI in each bin. The patterns suggest that the relationship is not uniformly linear and that its shape differs across the response distribution. These descriptive comparisons motivate a model that accommodates both sources of variation.

To address these questions, we use a semiparametric quantile regression framework with natural cubic splines for NDVI. The spline component allows the association to vary across vegetation levels, while quantile regression allows it to differ across the conditional response distribution. We apply this framework to satellite-derived observations within the Bruhat Bengaluru Mahanagara Palike (BBMP) boundary during 2000–2025, controlling for urban category and nighttime lights. Extended specifications include lagged LST and neighboring NDVI to examine temporal persistence and the additional association with surrounding vegetation. LST is deseasonalized separately for each pixel and calendar month, so the response represents deviations from local monthly temperature baselines. Accordingly, upper conditional quantiles describe relatively warm temperature anomalies rather than absolute-temperature extremes.

The study brings these components together to describe how local and surrounding vegetation are associated with the level and spread of LST anomalies. Finite-change contrasts summarize the nonlinear relationship on a common NDVI scale, while sensitivity analyses examine its dependence on model specification and seasonal conditions. The objective is to characterize adjusted associations rather than establish causal effects. The findings can inform discussions of urban heat mitigation by identifying vegetation levels and parts of the temperature-anomaly distribution where the estimated associations are strongest.

The remainder of this paper is organized as follows. In Section 2, we describe the satellite-derived remote sensing data used in this paper, along with the necessary pre-processing steps and some relevant

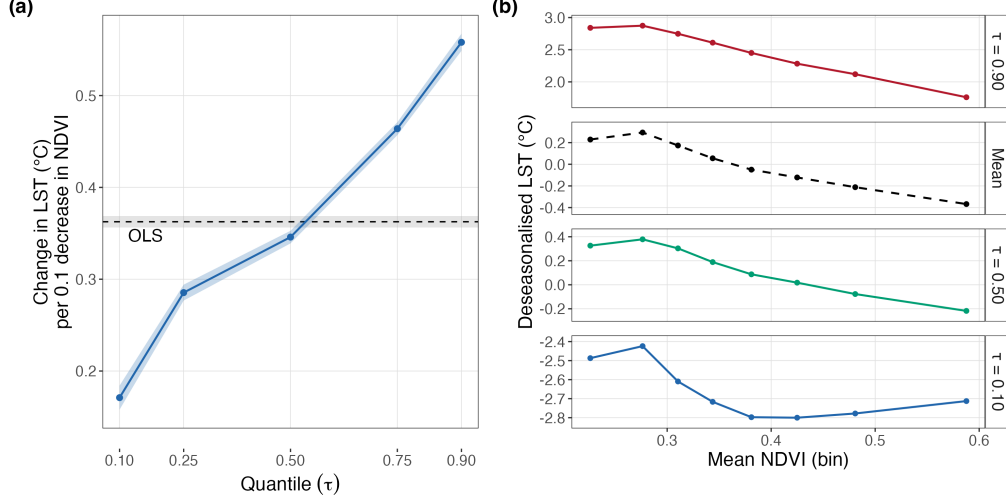


Figure 1: Motivation for a flexible quantile regression model. Panel (a) compares estimated associations for a 0.1-unit decrease in NDVI across quantile levels with the OLS estimate. Panel (b) shows the mean and selected quantiles of deseasonalized LST within eight NDVI bins, plotted against the mean NDVI in each bin.

exploratory data analysis. Section 3 presents our methodology and modeling framework, while the main results are illustrated in Section 4. We conclude the paper with some important remarks and directions for future research in Section 5.

2. Data and exploratory analysis

The study focuses on Bengaluru, the capital city of the state of Karnataka in India, for the time period 2000 until 2025. We consider the Bruhat Bengaluru Mahanagara Palike (BBMP) administrative boundary, covering approximately 716 km². This analysis combines data of various remote sensed variables such as land surface temperature (LST), normalized difference vegetation index (NDVI), urban land cover (ULC) and nighttime light (NTL).

The main response variable for the study is LST, which is obtained from the MODIS/Terra Land Surface Temperature and Emissivity product MOD11A2.061 (Wan et al., 2021) for different pixels within the BBMP boundary. It provides 8-day composite daytime LST observations at 1 km spatial resolution. The data are downloaded using the NASA AppEEARS platform (link: <https://appeears.earthdatacloud.nasa.gov/>). The main regressor variable in our analysis is NDVI which quantifies the greenness or vegetation condition of the location. Numerically, it is recorded in the interval $(-1, 1)$, with higher positive values indicating greener areas whereas values near to or less than 0 are generally water bodies, rocks, sand etc. NDVI observations in our study are obtained from the MODIS/Terra Vegetation Indices product MOD13Q1.061 (Didan, 2021), which provides observations at 250 m spatial resolution and 16-day temporal resolution. This 16-day NDVI composite window is also used as the reference temporal unit for the entire analysis.

Next, data on ULC are obtained from the MODIS Land Cover Type product MCD12Q1.061 (Friedl and Sulla-Menashe, 2019). This provides data at an annual temporal resolution, and since land cover does not change rapidly within a year, for the 16-day composite we consider the same value throughout the year. This product provides global land cover classifications at 500 m spatial resolution. The IGBP (International Geosphere-Biosphere Programme) classification layer, LC Type1, is used to identify urban and built-up land. In this classification, urban and built-up land corresponds to class 13. For each 1 km grid cell, the fraction of 500 m sub-pixels classified as urban is computed. This produces a continuous urban fraction variable ranging from 0 to 1. A categorical urban variable is subsequently

constructed: grid cells with urban fraction equal to 0 are classified as non-urban, grid cells with urban fraction between 0 and 1 are classified as mixed, and grid cells with urban fraction equal to 1 are classified as fully urban. Since MCD12Q1.061 is available annually from 2001 to 2023 for the study period, the 2001 land cover classification is assigned to the year 2000, while the 2023 classification is carried forward for 2024 and 2025. Finally, the NTL variable is used as a proxy for economical and other anthropogenic activities that denotes urban intensity. We obtained this data from the HKU DataHub (Cheng et al., 2026, link: <https://datahub.hku.hk/>). This provides a 500m spatial resolution with monthly temporal granularity. We note that several studies such as Li et al. (2019) have established a positive trend between NTL and LST, thereby motivating its use in our study as an additional control.

All datasets are spatially and temporally combined to construct the final spatio-temporal panel. Spatially, all variables are structured appropriately to the common 1 km grid defined by the LST MOD11A2.061 pixels within the BBMP boundary. Temporally, the 16-day MOD13Q1 NDVI composite window is used as the reference period. The 8-day LST observations falling within each 16-day NDVI window are averaged. As mentioned earlier, ULC observations are assigned to all 16-day windows within the corresponding year, while monthly NTL observations are assigned to all 16-day windows falling within the corresponding month. The complete dataset consists of 955 unique 1 km grid cells observed over 579 sixteen-day periods during 2000–2025. After excluding observations with missing LST values (due to cloud cover and other retrieval-related issues), the final dataset contains 4,79,413 observations. This panel structure enables the analysis of the nonlinear relationship between vegetation and land surface temperature while controlling for urban land cover and nighttime light intensity.

To better understand the data, we begin with a brief exploratory analysis. Table 1 presents descriptive statistics for land surface temperature (LST) across years and pixels. LST exhibits a pronounced seasonal pattern, with higher temperatures during the summer months (February–May) and relatively lower temperatures during the monsoon (June–October) and winter (November–January). To remove this systematic seasonal variation, we deseasonalize LST separately for each pixel and calendar month which serves as the response variable throughout the analysis.

Table 1: Monthly distribution of land surface temperature across pixels and years, 2000–2025. All values are in °C; quartiles are reported as (Q1, median, Q3).

Month	Mean $\pm$ SD	Range	Quartiles
January	29.26 $\pm$ 1.88	14.25–34.98	(28.22, 29.33, 30.52)
February	33.49 $\pm$ 2.28	25.42–40.14	(31.89, 33.36, 35.08)
March	36.37 $\pm$ 2.39	25.38–42.88	(34.87, 36.61, 38.05)
April	36.09 $\pm$ 2.59	28.40–44.15	(34.23, 36.21, 37.97)
May	32.83 $\pm$ 2.52	20.69–42.79	(31.30, 32.88, 34.46)
June	30.52 $\pm$ 2.63	13.49–40.11	(28.81, 30.50, 32.23)
July	29.82 $\pm$ 2.58	18.67–38.85	(28.21, 29.77, 31.57)
August	30.49 $\pm$ 2.27	17.59–38.77	(29.12, 30.55, 31.96)
September	30.68 $\pm$ 2.51	16.69–40.23	(29.11, 30.59, 32.24)
October	29.48 $\pm$ 2.09	18.69–36.97	(28.06, 29.48, 30.99)
November	28.50 $\pm$ 2.02	14.69–34.67	(27.43, 28.69, 29.77)
December	27.87 $\pm$ 1.73	19.48–33.25	(27.07, 28.12, 28.99)

Next, in Figure 2, we show the spatial variation of the four variables considered for the study in the four panels. All observations are averaged for the entire timeline. The spatial distribution of these variables exhibits substantial heterogeneity across the study area. From the top two panels, we can observe an inverse relationship between NDVI and LST: areas with relatively higher NDVI generally have lower LST values, whereas areas with lower vegetation exhibits higher surface temperatures. This

inverse relationship is also reflected in the correlation between the two variables, with LST and NDVI exhibiting a moderate negative correlation ($r = -0.47$, $p < 0.001$). The spatial contrast is particularly evident around $12.95^\circ\text{--}13.05^\circ\text{N}$ and $77.50^\circ\text{--}77.60^\circ\text{E}$, where relatively lower NDVI values coincide with higher LST, while comparatively higher NDVI and lower LST values are visible around $12.90^\circ\text{--}12.95^\circ\text{N}$ and $77.65^\circ\text{--}77.75^\circ\text{E}$. Nighttime lights also show a similar inverse relationship with LST. The urban land-cover classification shows that much of the central region is predominantly urban, surrounded by a mix of mixed and non-urban pixels, consistent with the spatial expansion of built-up areas associated with rapid urbanization in Bengaluru. However, the areas with relatively high NDVI and lower LST are predominantly classified as non-urban, particularly around $12.90^\circ\text{--}12.95^\circ\text{N}$ and $77.65^\circ\text{--}77.8^\circ\text{E}$.

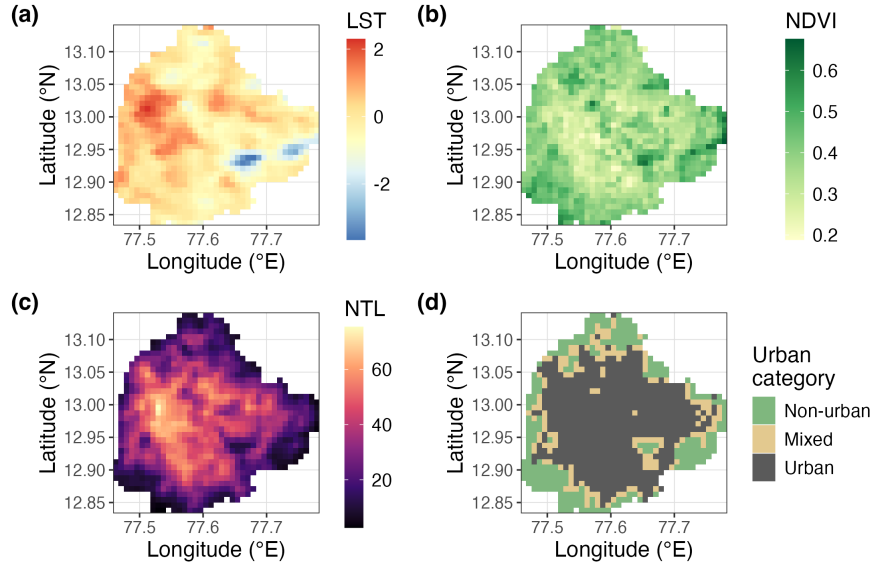


Figure 2: Average LST (a), NDVI (b), nighttime lights (c) and urban category (d) across years

The descriptive statistics of the variables are as follows: NDVI has a mean of 0.38 and ranges from -0.08 to 0.94 , reflecting variation in vegetation cover across the study area. For urban category, majority of the pixels are urban. In percentage terms, 25.4% of the observations are non-urban, 16.8% are mixed while 57.6% are fully urban. NTL shows considerable variability, with a mean of 32.07 and values ranging from 0 to 162.89, reflecting differences in nighttime activity and urban intensity.

3. Methodology

We start by setting some important notations. Throughout this article, we use $\mathbb{I}(A)$ as the indicator function which equals one when the condition A holds and zero otherwise. Unless otherwise specified, $i = 1, \dots, n$ is used to index pixels while t index the 16-day observation periods. For a random variable Z , use $Q_\tau(Z | \mathcal{C})$ to define its conditional quantile at probability level $\tau \in (0, 1)$ given covariates $\mathcal{C}$. In all our illustrations, we consider five different quantile levels, namely $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$.

3.1. Semiparametric conditional quantile model

The analysis examines whether the association between vegetation and land surface temperature (LST) varies across vegetation levels and across the conditional temperature distribution. Quantile regression allows these associations to differ between the lower tail, center and upper tail of the response distribution. A natural cubic spline represents the potentially nonlinear association with local normalized difference vegetation index (NDVI), while the remaining covariates enter parametrically. We therefore refer to the specification as a semiparametric quantile regression model.

As discussed in section 2, LST is deseasonalized separately for each pixel and calendar month. Let T_{it} denote observed LST and let $m(t)$ identify the calendar month assigned to period t . The response is

$$Y_{it} = T_{it} - \bar{T}_{i,m(t)}, \quad (1)$$

where $\bar{T}_{i,m}$ is the mean of the available LST observations for pixel i in calendar month m across years. The pixel's overall mean is not added back. Consequently, Y_{it} measures a temperature anomaly relative to that pixel's monthly baseline, in degrees Celsius. The transformation removes pixel-specific monthly mean differences; it does not remove long-term trends or all temporal dependence. Upper conditional quantiles describe relatively warm anomalies after conditioning on the model covariates, rather than the highest absolute temperatures in the study area.

Let X_{it} denote local NDVI, N_{it} nighttime light intensity and U_{it} urban category. We code $U_{it} = 0$ for non-urban pixels, $U_{it} = 1$ for mixed pixels and $U_{it} = 2$ for fully urban pixels. The full specification is

$$Q_\tau(Y_{it} | \mathcal{C}_{it}) = \alpha_\tau + \rho_\tau Y_{i,t-1} + f_\tau(X_{it}) + \delta_\tau X_{it}^{(W)} + \beta_{1\tau} \mathbb{I}(U_{it} = 1) + \beta_{2\tau} \mathbb{I}(U_{it} = 2) + \gamma_\tau N_{it}, \quad (2)$$

where $\mathcal{C}_{it}$ contains the covariates on the right-hand side. All estimated coefficients are allowed to vary with τ . The coefficient ρ_τ describes the association with LST in the preceding 16-day period. Here, $t - 1$ denotes the preceding scheduled period, rather than the most recent available observation when periods are missing. The urban-category coefficients compare conditional quantiles of temperature anomalies with those for the non-urban reference category; they do not estimate differences in absolute LST between urban categories. The local NDVI term is

$$f_\tau(x) = \sum_{k=1}^K \theta_{k\tau} B_k(x), \quad (3)$$

where B_k denotes a natural cubic spline basis function. We use $K = 4$ basis functions, with internal knots selected from the empirical NDVI distribution. A sensitivity analysis about this choice is carried out as well. The basis excludes an intercept because the model includes α_τ . Natural cubic splines are piecewise cubic and linear beyond their boundary knots. The same basis specification is used across quantiles, allowing the fitted coefficients to describe differences in the NDVI association.

To represent surrounding vegetation, let $\mathcal{N}_i^{(r)}$ denote the set of r nearest neighboring pixels, excluding pixel i . We define $V_i(t)$ as the subset with observed NDVI at time t , and let $v_i(t) = |V_i(t)|$. The neighborhood covariate is

$$X_{it}^{(W)} = \sum_{j \in V_i(t)} w_{ij}(t) X_{jt}, \quad w_{ij}(t) = \frac{1}{v_i(t)}, \quad j \in V_i(t), \quad (4)$$

with zero weights for pixels outside $V_i(t)$. The average is defined when $v_i(t) > 0$. The primary specification uses $r = 8$; the number of observed neighbors can be smaller when NDVI is missing. The coefficient δ_τ describes the association with neighboring vegetation conditional on local NDVI and the other covariates. Including this covariate does not itself eliminate residual spatial dependence.

3.2. Estimation and finite-change contrasts

For each quantile, coefficients are estimated by minimizing the check loss, which is defined as

$$\ell_\tau(u) = u\{\tau - \mathbb{I}(u < 0)\} \quad (5)$$

for quantile level τ . Let $\mathbf{z}_{it}$ collect the intercept, lagged response, spline basis values, neighborhood NDVI, urban-category indicators and nighttime lights, and let $\boldsymbol{\eta}_\tau$ collect their coefficients in the same order. For the set $\mathcal{A}$ of observations with the response and all covariates required by a specification, the estimator is

$$\hat{\boldsymbol{\eta}}_\tau = \arg \min_{\boldsymbol{\eta}_\tau} \sum_{(i,t) \in \mathcal{A}} \ell_\tau(Y_{it} - \mathbf{z}'_{it}\boldsymbol{\eta}_\tau). \quad (6)$$

To summarize the nonlinear NDVI association on an interpretable scale, we calculate a finite-change contrast for a 0.1-unit decrease in local NDVI. Let $\mathbf{c}$ denote fixed values of the remaining covariates. The estimated contrast at starting value x is

$$\hat{\Delta}_\tau(x) = \hat{Q}_\tau(Y \mid X = x - 0.1, \mathbf{c}) - \hat{Q}_\tau(Y \mid X = x, \mathbf{c}). \quad (7)$$

Note that, within a given fitted model, the above contrast does not depend on the fixed values chosen for the other covariates. It is a finite difference, rather than a derivative. Positive values indicate that lower NDVI is associated with a higher conditional quantile of the temperature anomaly; negative values indicate the reverse. Both endpoints must lie within the NDVI range over which the fitted relationship is evaluated. Contrasts from the full model hold lagged LST and neighborhood NDVI fixed and are not long-run responses to a persistent change in vegetation.

Uncertainty in the contrasts is summarized using a pixel-level cluster bootstrap. In each of B replications, pixels are sampled with replacement and all available observations from each selected pixel are retained together. The model is refitted and the contrast in (7) is recomputed. If $\hat{\Delta}_\tau^{(b)}(x)$ denotes the contrast in replication b , the pointwise 95% percentile interval is

$$\left[q_{0.025} \left\{ \hat{\Delta}_\tau^{(b)}(x) : b = 1, \dots, B \right\}, q_{0.975} \left\{ \hat{\Delta}_\tau^{(b)}(x) : b = 1, \dots, B \right\} \right], \quad (8)$$

where q_p is the empirical quantile at probability p . Resampling whole pixels preserves the within-pixel temporal structure of the observed records. It does not preserve dependence between different pixels, so the intervals do not account fully for residual spatial dependence or common temporal shocks.

3.3. Alternative specifications and sensitivity analyses

To understand how the modeling choices influence our findings, we carry out a series of sensitivity analyses. We consider different sets of control variables, degrees of freedom for the spline, numbers of neighboring pixels, and temporal lags of NDVI. We also repeat the analysis after excluding monsoon months. These comparisons allow us to examine whether the main findings remain consistent across the specifications and seasonal conditions considered.

The first exercise is to compare the proposed model (2) with three reduced specifications. The baseline model includes local NDVI, urban category and nighttime lights:

$$Q_\tau \left(Y_{it} \mid \mathbf{c}_{it}^{(0)} \right) = \alpha_\tau + f_\tau(X_{it}) + \beta_{1\tau}\mathbb{I}(U_{it} = 1) + \beta_{2\tau}\mathbb{I}(U_{it} = 2) + \gamma_\tau N_{it}. \quad (9)$$

Then, we extend it to add surrounding NDVI:

$$Q_\tau \left(Y_{it} \mid \mathbf{c}_{it}^{(W)} \right) = \alpha_\tau + f_\tau(X_{it}) + \delta_\tau X_{it}^{(W)} + \beta_{1\tau}\mathbb{I}(U_{it} = 1) + \beta_{2\tau}\mathbb{I}(U_{it} = 2) + \gamma_\tau N_{it}. \quad (10)$$

Finally, we consider the temporal extension instead of the spatial one, by adding the preceding LST deviation:

$$Q_\tau \left(Y_{it} \mid \mathcal{C}_{it}^{(L)} \right) = \alpha_\tau + \rho_\tau Y_{i,t-1} + f_\tau(X_{it}) + \beta_{1\tau} \mathbb{I}(U_{it} = 1) + \beta_{2\tau} \mathbb{I}(U_{it} = 2) + \gamma_\tau N_{it}. \quad (11)$$

The conditioning sets contain the covariates included in their respective models. These specifications distinguish the local NDVI association from associations conditional additionally on surrounding vegetation, recent temperature anomalies, or both.

We also assess sensitivity to spline flexibility using $K \in \{2, 3, 4, 5, 6\}$ and to the neighborhood definition using $r \in \{4, 8, 12, 16\}$. The four-basis-function specification provides the principal summary of the nonlinear association; the alternatives assess whether additional flexibility materially improves the fit. Another robustness study is done to examine the impact of NDVI lags $s \in \{0, 1, 2, 4, 8, 12, 23\}$, where $s = 0$ denotes contemporaneous NDVI and reflects the model in (2). For other choices of s , we replace the X_{it} term in the model with $X_{i,t-s}$ and evaluate its performance. Observe that the lag $s = 1$ corresponds to approximately 16 days, $s = 12$ to approximately six months and $s = 23$ to approximately one year. These comparisons concern NDVI levels at different times, rather than temporal changes in NDVI. As the last exercise, we repeat the entire analysis after excluding monsoon months to assess sensitivity to seasonal conditions and the availability of cloud-free satellite retrievals.

Note that in all explorations, the fit of the model is summarized using the mean check loss on an evaluation set $\mathcal{E}$:

$$\bar{L}_\tau(\mathcal{E}) = \frac{1}{|\mathcal{E}|} \sum_{(i,t) \in \mathcal{E}} \ell_\tau \left(Y_{it} - \hat{Q}_\tau(Y_{it} \mid \mathcal{C}_{it}) \right). \quad (12)$$

At a given quantile, lower loss indicates better performance on the evaluation observations. Out-of-sample performance was assessed using a temporal holdout design. The models were trained on observations from 2000-2020 and tested on observations from 2021-2025. A temporal split was preferred to a random train-test partition because the data consist of repeated observations over the same spatial pixels, and random splitting could lead to leakage across adjacent time periods. For the evaluation, a common sample was considered across model specifications.

3.4. Spatial scale of the neighborhood association

We examine how the estimated neighborhood association changes when surrounding vegetation is averaged using alternative distance-decay scales. Let d_{ij} be the Haversine distance, in kilometers, between the centroids of pixels i and j . For a decay scale $\lambda > 0$, define the unnormalized weight

$$a_{ij}(\lambda) = \exp(-d_{ij}/\lambda), \quad j \neq i. \quad (13)$$

If $\mathcal{O}_i(t)$ denotes the other pixels in the study domain with observed NDVI at time t , the distance-weighted covariate is

$$X_{it}^{(W,\lambda)} = \frac{\sum_{j \in \mathcal{O}_i(t)} a_{ij}(\lambda) X_{jt}}{\sum_{j \in \mathcal{O}_i(t)} a_{ij}(\lambda)}. \quad (14)$$

The weights are normalized over the available neighboring observations. For each $\lambda \in \{0.5, 1, 2, 3, 5, 10\}$ km, we refit (10), replacing $X_{it}^{(W)}$ with $X_{it}^{(W,\lambda)}$. These models include local NDVI, urban category and nighttime lights, but not lagged LST. Writing the resulting coefficient as $\delta_\tau(\lambda)$, the estimated contrast for a 0.1-unit increase in weighted neighboring NDVI is $0.1\hat{\delta}_\tau(\lambda)$. The distance-decay analysis is used as a spatial-scale sensitivity check. By varying λ , we examine whether the neighboring vegetation association is driven mainly by very nearby pixels or by a broader surrounding vegetation field. The results should therefore be interpreted as evidence on the spatial scale at which neighboring greenness is most strongly associated with local LST, rather than as an estimate of a fixed cooling distance.

4. Results and discussion

For the discussions below, we label the four model specifications as follows: the full model with lagged LST and neighborhood NDVI (2) is denoted by M1, the baseline model in (9) by M2, the neighborhood NDVI model in (10) by M3, and the autoregressive LST model in (11) by M4.

Table 2 compares the four specifications across the five quantiles using mean quantile check loss. In-sample, the full model M1 reports the lowest loss at every quantile, indicating the best fit among the candidate specifications. Out-of-sample, most of the improvement over the baseline model M2 is obtained after including lagged LST. The autoregressive model M4 performs slightly better than M1 at $\tau = 0.10$ and $\tau = 0.25$, the two models are tied at the reported precision for $\tau = 0.50$ and $\tau = 0.75$, while M1 performs slightly better at $\tau = 0.90$. In contrast, adding neighborhood NDVI alone in M3 produces only small improvements relative to the baseline. These results suggest that recent temperature anomalies contain substantial predictive information, whereas neighborhood NDVI contributes comparatively less to predictive performance.

Table 2: In-sample and out-of-sample mean quantile check loss across model specifications.

Model	$\tau = 0.10$		$\tau = 0.25$		$\tau = 0.50$		$\tau = 0.75$		$\tau = 0.90$	
	In-sample	Out-of-sample	In-sample	Out-of-sample	In-sample	Out-of-sample	In-sample	Out-of-sample	In-sample	Out-of-sample
M1	0.3626	0.555	0.6297	0.934	0.7563	1.030	0.5811	0.730	0.3157	0.375
M4	0.3629	0.551	0.6303	0.931	0.7582	1.030	0.5846	0.730	0.3179	0.377
M3	0.3931	0.694	0.6630	1.050	0.7804	1.110	0.5970	0.780	0.3261	0.404
M2	0.3938	0.696	0.6654	1.050	0.7841	1.110	0.6021	0.784	0.3293	0.408

However, predictive improvement and environmental interpretation address different questions. Although the neighborhood NDVI term adds only modest predictive gains, it remains useful for describing the association between local LST and surrounding vegetation conditions. Therefore, we use the baseline model M2 to present the local NDVI-LST relationship and the corresponding finite-change contrasts, while the neighborhood vegetation associations are reported using both M3 and the full model M1, with and without adjustment for lagged LST.

Figure 3 shows the fitted conditional quantiles for NDVI between 0.30 and 0.65. Nighttime lights are fixed at their sample mean and urban category is set to mixed. The median declines from approximately 0.6°C to -0.6°C over this range. The upper quantiles also decline, with the 90th percentile decreasing from approximately 3.0°C to 1.1°C . The 10th percentile decreases initially and is approximately flat beyond NDVI 0.40, with a small intervening increase. Thus, the relationship is neither linear nor a uniform downward shift of all conditional quantiles.

The fitted 10th–90th percentile range narrows from approximately 5.4°C at NDVI 0.30 to 4.1°C at NDVI 0.65. This pattern indicates lower conditional dispersion in temperature anomalies at higher NDVI, driven mainly by the decline in the upper quantiles over the moderate-to-dense vegetation range. A conditional mean model alone would not describe these differences between the upper and lower parts of the distribution. The descriptive sparse, moderate and dense vegetation groups have mean urban fractions of approximately 79%, 48% and 9%, respectively. These summaries illustrate the overlap between vegetation and urbanization in the observed data; they are not the urban fractions used to generate the fitted curves, for which urban category is held fixed. The NDVI group boundaries are descriptive and should not be treated as independently estimated ecological thresholds. The patterns are compatible with vegetation-related shading and evapotranspiration, but we do not establish causal mechanisms.

Figure 4 presents four finite-change contrasts. A 0.1-unit decrease provides a common scale for comparing the nonlinear association at different starting NDVI levels. For the change from 0.35 to 0.25, all five estimated contrasts are positive, approximately 0.49 – 0.61°C , and the displayed intervals exclude zero. The association is therefore positive throughout the examined conditional distribution in

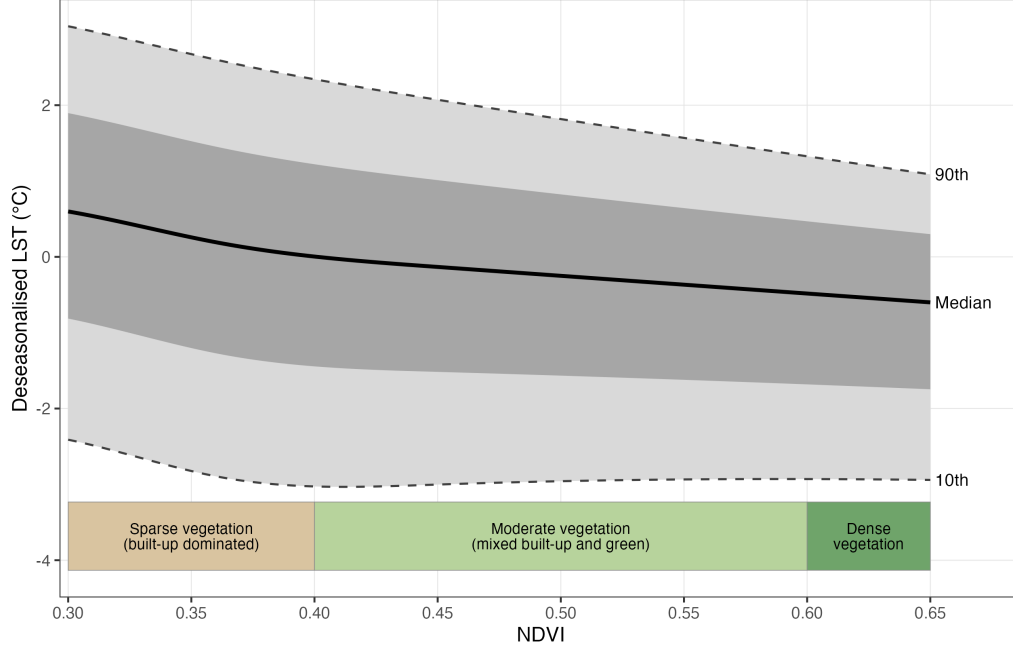


Figure 3: Fitted conditional quantiles of LST anomalies from the baseline model as a function of local NDVI, with nighttime lights fixed at their sample mean and urban category set to mixed. The solid curve is the median; the dark and light bands span the 25th–75th and 10th–90th fitted quantiles, respectively. These bands describe conditional dispersion, not estimation uncertainty. The vegetation labels are descriptive NDVI groupings.

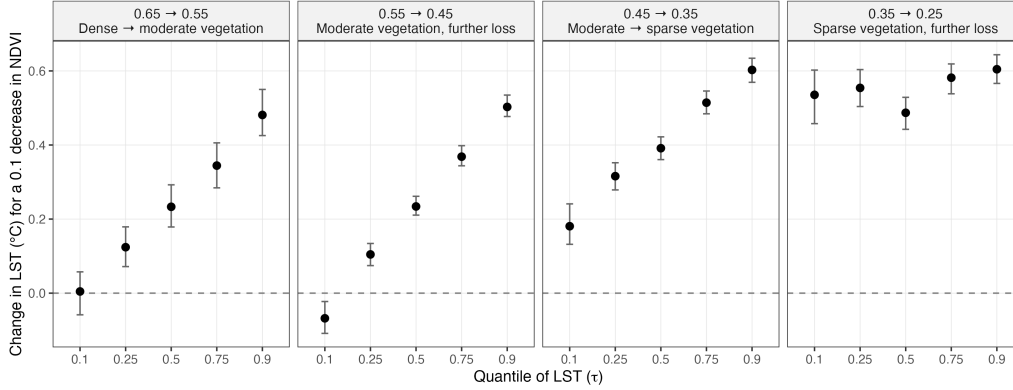


Figure 4: Estimated changes in conditional quantiles of LST anomalies associated with a 0.1-unit decrease in local NDVI, from the baseline model. Panels identify the starting and ending NDVI values. Points are finite-change estimates and bars are pointwise 95% pixel-cluster bootstrap intervals. The dashed line denotes no change.

this low-NDVI range. Although the intervals overlap across quantiles, overlap alone does not establish equality of the contrasts.

For the change from 0.45 to 0.35, the estimated increase rises from approximately 0.18°C at $\tau = 0.10$ to 0.60°C at $\tau = 0.90$, with a median contrast of about 0.39°C . This quantile gradient indicates a stronger association in the upper part of the conditional anomaly distribution. It is consistent with the narrowing of the fitted quantile range as NDVI increases through this region.

For the changes from 0.65 to 0.55 and from 0.55 to 0.45, median contrasts are approximately 0.23°C , while 90th-percentile contrasts are approximately 0.48°C and 0.50°C , respectively. The corresponding 10th-percentile contrasts are approximately 0.01°C and -0.07°C . The interval for the former includes zero, whereas the displayed interval for the latter lies below zero. These results show

that the NDVI association can differ in both magnitude and direction across quantiles. They do not support a uniformly positive contrast at all NDVI levels and quantiles.

Table 3 summarizes the finite-change curve over the evaluation grid. The largest estimated contrasts occur at starting NDVI values between approximately 0.38 and 0.39; the largest reported value is 0.7010°C at $\tau = 0.90$. Because the contrast compares x with $x - 0.1$, these locations refer to the starting values of finite intervals, rather than pointwise derivative maxima or independently identified land-cover thresholds. The mean and median contrasts in the table summarize the evaluated grid values and are not necessarily averages over the observed NDVI distribution.

We additionally summarize the contrast curve using a grid-based flattening point. For an ordered evaluation grid $x_1 < \dots < x_G$, this is the first eligible x_g satisfying

$$\left| \hat{\Delta}_{\tau}(x_g) - \hat{\Delta}_{\tau}(x_{g-1}) \right| \leq \epsilon, \quad \epsilon = 0.001^{\circ}\text{C}. \quad (15)$$

The reported points lie near NDVI 0.52–0.53 for $\tau = 0.25$ through 0.90, with no point reported for $\tau = 0.10$. This criterion is sensitive to the grid spacing and search domain. Moreover, a nearly constant finite-change contrast can remain appreciably different from zero: at $\tau = 0.90$, the reported contrast at the flattening point is 0.5106°C . A constant finite difference can also arise in a linear segment of a natural spline. These values are therefore descriptive features of the fitted contrast curve, rather than evidence that vegetation-related cooling has ceased or reached a physical saturation threshold.

Table 3: Finite-change contrasts for a 0.1-unit decrease in local NDVI, summarized over the evaluation grid. Contrasts are in degrees Celsius; flattening points follow the grid-based rule in (15). NA denotes no reported point.

τ	Flattening NDVI	Contrast at flattening	Maximum contrast	Mean contrast	Median contrast	NDVI at maximum contrast
0.10	NA	NA	0.6765	0.1662	0.0008	0.3802
0.25	0.5217	0.1034	0.6687	0.2820	0.1555	0.3802
0.50	0.5217	0.2378	0.6086	0.3395	0.2588	0.3871
0.75	0.5310	0.3745	0.6839	0.4588	0.4007	0.3871
0.90	0.5240	0.5106	0.7010	0.5503	0.5200	0.3941

Table 4 reports contrasts for a 0.1-unit increase in the average NDVI of the eight nearest neighbors. Without lagged LST, the estimated contrasts range from -0.3000°C at $\tau = 0.10$ to -0.5300°C at $\tau = 0.90$. With lagged LST included, they remain negative and range from -0.1771°C to -0.4280°C . Both specifications show larger absolute associations at the upper quantiles. The attenuation after adjustment for lagged LST is compatible with overlapping information in neighborhood vegetation and recent temperature anomalies.

Table 4: Estimated contrasts for a 0.1-unit increase in eight-neighbor mean NDVI, in degrees Celsius.

τ	Contrast for a 0.1-unit increase in $X^{(W)}$	
	Full model	Baseline + neighborhood NDVI
0.10	-0.1771	-0.3000
0.25	-0.1905	-0.3700
0.50	-0.2798	-0.3900
0.75	-0.4104	-0.4900
0.90	-0.4280	-0.5300

4.1. Sensitivity analyses

To understand how the modeling choices influence our findings, we examine different numbers of neighboring pixels, levels of spline flexibility, and temporal lags of NDVI. Table 5 presents the corresponding mean check losses in three panels, with lower values indicating better fit at a given quantile. We also repeat the analysis after excluding monsoon months to examine whether the estimated associations remain consistent under different seasonal conditions.

Panel (a) in table 5 shows little variation in mean check loss when the number of neighbors increases from 4 to 16. Although twelve neighbors yield the lowest reported loss at each quantile, the largest difference within a quantile is approximately 0.0007. The eight-neighbor specification therefore provides a similar fit to the alternatives. This comparison indicates limited sensitivity of model fit to neighborhood size.

Panel (b) compares different numbers of natural spline basis functions. Increasing flexibility beyond four basis functions produces no improvement at the reported precision, while using two basis functions gives slightly higher loss at the lower quantiles and the median. These results support four basis functions as a parsimonious representation of the nonlinear relationship.

Panel (c) compares alternative temporal lags of NDVI. Contemporaneous NDVI gives the lowest reported loss at $\tau = 0.25, 0.50, 0.75$, and 0.90 , while lags 1 and 12 perform marginally better at $\tau = 0.10$. For most of the quantiles, since contemporaneous NDVI provides at par or lesser losses, it suggests current vegetation conditions are more informative for these parts of the response distribution.

Finally, excluding monsoon months yielded qualitatively similar estimates (results not shown in the interest of space). This suggests that the main patterns are not confined to monsoon conditions, although the magnitude and precision of any differences require further comparison.

Table 5: Mean check loss across alternative neighborhood sizes, spline specifications, and NDVI lags. Here, r is the number of nearest neighbors, K is the number of natural spline basis functions, and s is the NDVI lag in 16-day periods.

Specification	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$
Panel (a): Number of nearest neighbors					
$r = 4$	0.39315	0.66307	0.78044	0.59724	0.32618
$r = 8$	0.39302	0.66274	0.78009	0.59675	0.32594
$r = 12$	0.39294	0.66257	0.77992	0.59654	0.32585
$r = 16$	0.39297	0.66262	0.78000	0.59665	0.32597
Panel (b): Spline flexibility					
$K = 2$	0.395	0.667	0.785	0.602	0.329
$K = 3$	0.394	0.666	0.784	0.602	0.329
$K = 4$	0.394	0.665	0.784	0.602	0.329
$K = 5$	0.394	0.665	0.784	0.602	0.329
$K = 6$	0.394	0.665	0.784	0.602	0.329
Panel (c): Temporal lag of NDVI					
$s = 0$	0.395	0.668	0.787	0.603	0.329
$s = 1$	0.394	0.669	0.790	0.606	0.331
$s = 2$	0.395	0.672	0.793	0.608	0.332
$s = 4$	0.395	0.673	0.795	0.611	0.333
$s = 8$	0.396	0.672	0.797	0.620	0.342
$s = 12$	0.394	0.672	0.797	0.620	0.342
$s = 23$	0.395	0.673	0.798	0.618	0.339

4.2. Sensitivity to the spatial decay scale

Figure 5 shows the estimated contrast in conditional LST anomalies associated with a 0.1-unit increase in exponentially weighted neighboring NDVI across the candidate decay scales. All point estimates are negative, indicating that higher surrounding greenness is associated with lower local LST anomalies across all quantiles and decay scales. At $\lambda = 0.5$ km, the estimated contrasts range approximately from -0.39°C to -0.55°C , while at $\lambda = 10$ km they range approximately from -0.24°C to -0.45°C . The magnitude of the association generally weakens as the weighting becomes more spatially diffuse, although the pattern is not strictly monotone for every quantile.

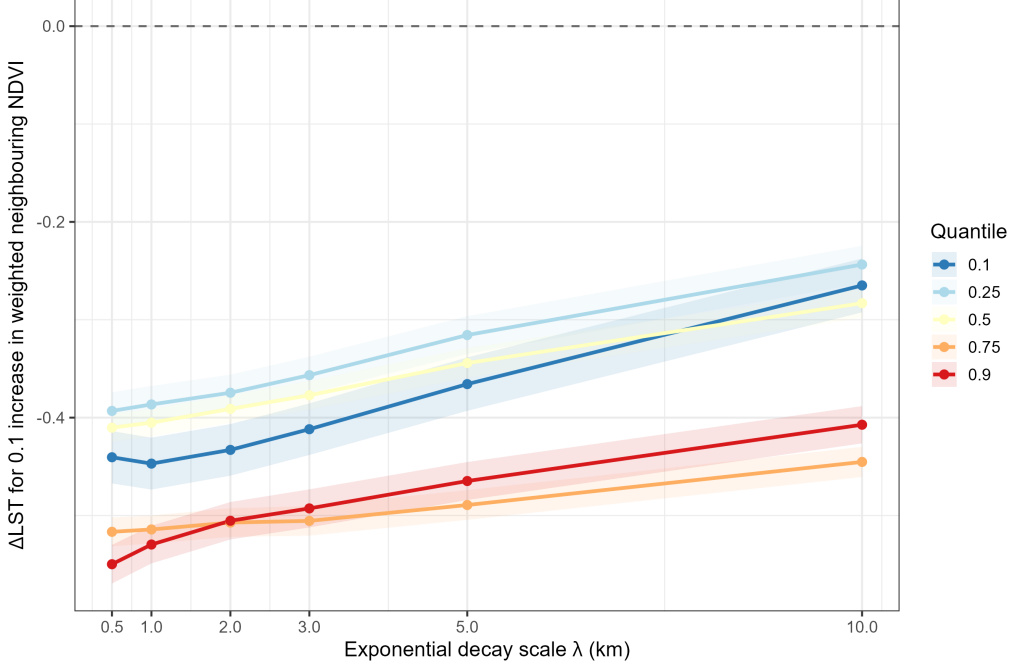


Figure 5: Estimated changes in conditional quantiles of LST anomalies associated with a 0.1-unit increase in exponentially weighted neighboring NDVI, across spatial decay scales. Models adjust for local NDVI, urban category and nighttime lights, and exclude lagged LST. Lines connect estimates from separately fitted models; shading reproduces the uncertainty bands supplied with the figure. The decay scale controls spatial weighting and is not a maximum cooling distance.

The upper quantiles generally show stronger negative associations than the lower quantiles and the median, suggesting that neighboring vegetation is more strongly associated with reductions in hotter surface-temperature conditions. However, the ordering of the upper quantiles changes with the decay scale: the 90th-percentile contrast is more negative at the smallest decay scales, whereas the 75th-percentile contrast is more negative at larger decay scales. Thus, the figure supports a stronger association in the upper part of the conditional LST distribution, but not a uniform ranking between the two upper quantiles.

These results indicate that the estimated neighboring-vegetation association depends on how surrounding greenness is spatially summarized. The decay scale λ controls the distance weighting used to construct neighboring NDVI; it should not be interpreted as a maximum cooling distance. Each estimate comes from a separate model using a different weighted NDVI covariate, and each weighted average includes contributions from multiple distances. Therefore, a negative estimate at $\lambda = 10$ km does not imply that vegetation has a physical cooling effect extending exactly 10 km. Rather, the results show that surrounding vegetation remains negatively associated with local LST anomalies under broader spatial weighting, within the observed study domain. These estimates should be interpreted as adjusted associations, not as causal effects of vegetation interventions.

5. Conclusions and future directions

There is a growing debate in environmental conservation and urban planning about the trade-off between development and environmental protection. In the early stages of urban growth, environmental degradation may appear to accompany economic development, and its short-run benefits may seem to outweigh its costs. However, the long-run consequences of environmental degradation are often nonlinear. Beyond a certain point, the environmental and social costs may increase sharply, making it important to identify possible thresholds or saturation points. In the context of urban heat, understanding such nonlinearities is especially relevant because vegetation loss may not affect land surface temperature at a constant rate across all greenness levels.

This study therefore examined the nonlinear and distributional relationship between NDVI and land surface temperature in Bangalore using a semi-parametric quantile regression framework. The results show that the NDVI-LST relationship is not adequately described by a single linear mean effect. Instead, the association varies across the NDVI range and across the conditional distribution of LST. The findings suggest that vegetation is particularly important under high-temperature conditions, as the association between NDVI loss and LST is stronger in the upper quantiles. The results also indicate the presence of threshold and saturation patterns, where the marginal temperature response to changes in vegetation differs across low, moderate, and high NDVI levels.

The study further shows that greener areas are not only associated with lower LST, but also with a narrower LST distribution. This suggests that vegetation may help reduce thermal variability and limit extreme-hot surface temperature outcomes. From an urban management perspective, these findings support the use of vegetation-based interventions as part of heat mitigation strategies. In particular, tree planting, green infrastructure development, and restoration of low-greenness areas may be more effective when targeted toward locations where vegetation loss is most strongly associated with higher LST.

We also considered the spatial aspect of vegetation cooling by incorporating a spatially lagged NDVI term. This allowed us to examine whether neighboring vegetation conditions are associated with local LST. The results suggest that vegetation in the surrounding area can contribute to local cooling, highlighting that urban greening strategies should not focus only on isolated pixels or individual green spaces, but also on the broader neighborhood vegetation structure.

Future research can extend this work in several directions. First, the modeling framework can be developed further to incorporate multiple responses, such as LST, air temperature, humidity, and heat stress indices, within a multivariate spatial quantile regression framework. Second, future work can extend the present approach to a fully spatio-temporal nonparametric quantile model to better capture temporal dependence, spatial dependence, and changing vegetation effects over time. Finally, future methodological work can focus on causal and policy-oriented extensions to assess how specific greening interventions may reduce extreme heat exposure in rapidly urbanising cities.

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